Time

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Time

How do I convert between different units of time?

- **Time** has a **number of different conversions** and does not use the decimal number system (based on 10s, 100s, etc)
  - You need to know the follow time conversions

Time conversions

There are...
... 60 seconds in a minute
... 60 minutes in an hour
... 24 hours in a day
... 7 days in a week
... 365 days in a year (366 in a leap year)
... 52 weeks in a year (plus one day)
... 12 months in a year
... 1000 years in a millennium

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- You should also know the **number of days** in each **calendar month**

Do learn these – and how many days are in each month...

Thirty days have September, April, June and November,
all the rest have thirty-one.
Excepting February alone.
And that has twenty-eight days clear,
and twenty-nine in each leap year.

January	31 days
February	28 (29 in a leap year)
March	31
April	30
May	31
June	30
July	31
August	31
September	30
October	31
November	30
December	31

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Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- A **12-hour clock** goes around once for AM and once for PM in **one complete day**
  - AM is between midnight (12 am) and midday (12 pm)
  - PM is between midday (12 pm) and midnight (12 am)

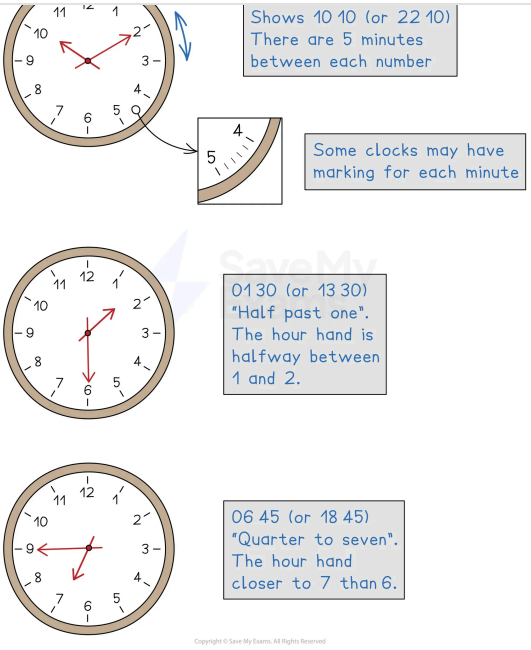
- A **24-hour clock** goes around once for **one complete day**
  - The display uses four digits, two for the hour, two for the minutes
  - The day starts at midnight which is 00 00 (12 am) and goes up to 23 59 (11.59 pm)

- You need to be able to **read the time** from both a **12-hour clock** and a **24-hour clock**

12-hour clock time	24-hour clock time
12:02am	00:02
9:37am	09:37
12:10pm	12:10
5:32pm	17:32
12:00am (Midnight)	00:00
12:00pm (Midday)	12:00

How do I read the time from an analogue clock?

- An **analogue** clock works in **12-hour** time
  - The **short hand** is the **hour hand**
    - Each number on the clock represents **one hour** for the hour hand
  - The **long hand** is the **minute hand**
    - Each number on the clock represents **five minutes** for the minute hand
    - Some clocks will have markings for individual minutes



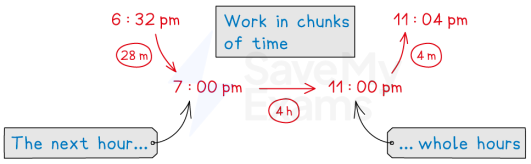
How do I read the time from a digital clock?

- A **digital** clock can use either **24 hour** time or **12-hour** time
  - A colon ':' is often displayed between the hours and minutes
    - E.g., 1245 would be displayed as 12:45
  - AM or PM does **not** need to be specified with 24-hour time
    - it may or may not be shown on a 12-hour time digital clock
  - For **single-digit** hours, clocks often miss out the first zero
    - e.g. 09:23 would be displayed as 9:23

How do I calculate with time using the 12-hour clock?

- Work in **chunks** of time
  - Calculate the **minutes until the next hour**, then **whole hours**, then **minutes until a final time**
- Ensure you know when the 12-hour clock **switches** from **AM to PM** and vice versa

and 11:04 pm, in hours and minutes.



4 h + 28 m + 4 m
4 h 32 m

There are 4 hours and 32 minutes between 6:32 pm and 11:04 pm.

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Examiner Tips and Tricks

Be careful when using a calculator, as they can often cause problems in time-based questions!

Remember that 1 hour 30 mins of time would be 1.5 hours on a calculator.

You can use the **degrees, minutes and seconds function** $\boxed{°'''}$ to enter time on your calculator

E.g. 3 minutes 45 seconds would be entered as $0°3'45''$

You can convert between an answer on the calculator, given in either format of time, by pressing this button.

Worked Example

(a) A film starts at 15 35 and lasts for 1 hour 45 minutes.

Find the time at which the film ends.

Answer:

Work in chunks of time
Add the hour on

15 35 + 1 hour = 16 35

Add on the number of minutes until the next hour
There are 60 minutes in an hour
Add on 25 minutes of the remaining 45 minutes to get to the next hour

16 35 + 25 minutes = 17 00

There are 45 - 25 = 20 minutes left of the film to add
Write your final answer using the type of clock used in the question (24-hour) unless otherwise stated

17 20

(b) It is due to start raining at 02 35 and not stop until 08 14.

Work in chunks of time
Find the number of minutes until the next hour

02 35 + 25 minutes = 03 00

Find the number of hours until the hour in the stopping time

03 00 + 5 hours = 08 00

Find the remaining minutes until the rain stops

08 00 + 14 minutes = 08 14

Add together the hours and minutes

25 minutes + 5 hours + 14 minutes

5 hours 39 minutes

(c) A train journey started at 11.23 am and finished at 2.27 pm.

How long, in hours and minutes, was the train journey?

Answer:

Work in chunks of time
Find the number of minutes until the next hour
Be careful about the AM/PM boundary

11.23 am + 37 minutes = 12.00 pm

Add on the number of hours until the hour in the arrival time

12.00 pm + 2 hours = 2.00 pm

Add on the remaining minutes until the arrival time

2.00 + 27 minutes = 2.27 pm

Add up the hours and minutes to find the total length of the train journey

37 minutes + 2 hours + 27 minutes

37 minutes + 27 minutes add up to 1 hour and 4 minutes

3 hours 4 minutes

How do I use bus and train timetables?

- Bus and train **timetables** tend to use the **24-hour** clock system
  - Each **column** represents a **different bus or train**
  - **Times** are listed as **four digits** without the colon ':'
  - The time in each cell usually indicates the **departure time** (when the bus/train leaves that stop/station)
  - The **last location** on the list usually shows the **arrival time**

Worked Example

The table below shows part of a bus timetable.

Coronation Street	0750	0800	0816
Albert Square	0818	0830	0849
Ramsey Street	0825	0840	0903
Emmerdale Village	0834	0852	0918

How long should the journey between Coronation Street and Ramsey street take on the 0800 bus?

Answer:

The 08 00 bus is the middle column

Find the time between the departure times at Coronation Street and Ramsey street for the middle column

08 40 - 08 00

40 minutes

How do I convert time between time zones?

- Different countries are in different **time zones**, depending on where they are on the planet
- For example when it is 11:36 in London in the UK, it may be 06:36 in New York in the USA
  - So New York is 5 hours **behind** London
  - London is 5 hours **ahead** of New York

- You must consider:
  - What is the **time difference** between the two locations (**in hours**)?
  - Which time zone is **ahead**, and which one is **behind**?

- For example, Buenos Aires (Argentina) is 6 hours behind Helsinki (Finland)
  - When it is 7 am in Helsinki
    - 7 am - 6 hours = 1 am in Buenos Aires

  - When it is 5 pm in Buenos Aires

Worked Example

The time in Madrid is 7 hours ahead of the time in Chicago.

Mark takes a flight from Madrid to Chicago which lasts 9 hours and 45 minutes.

The flight leaves at 7:30 am Madrid time.

What will the local time be in Chicago when he arrives?

Answer:

Work in Madrid time, and then change at the end

Add 9 hours 45 minutes on to 7:30 am to find the time Mark arrives in Chicago (in Madrid time)

07:30 + 9 h 45m = 16:30 + 45m = 17:15

17:15 is the time in Madrid, so find the time in Chicago

Madrid is 7 hours ahead of Chicago

Subtract 7 hours

17:15 - 7 hours = 10:15

10:15 am

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